SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks50** **Duration :60 Minutes**

Annexure A

Scenario-Based Questions

Code of Conduct:

I RUTHIKA REDDY C (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Marks (100)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Scenario-Based Questions

Question 1: Object Boundary Detection

(a) Interpret the scenario and explain the importance of boundary detection.
In medical images, tissue boundaries must be detected accurately for diagnosis.

(b) Identify key challenges in detecting boundaries.

- Image noise
- Low contrast
- Blurred edges
- Similar intensity values

(c) Apply a suitable edge detection method.
Canny Edge Detection

(d) Justify your selection with logical reasoning.
Canny detects clear, thin, and continuous edges while reducing noise.

(e) Present your explanation clearly and systematically.
Canny Edge Detection provides accurate tissue boundary detection.

Question 2: Simple Segmentation

(a) Interpret the scenario and explain segmentation requirements.
Objects are clearly different from the background based on intensity.

(b) Identify key features enabling separation.

- High contrast
- Different intensity values
- Uniform background

(c) Apply a suitable segmentation technique.
Threshold Segmentation

(d) Justify your decision logically.
Thresholding is simple, fast, and effective when object and background intensities are clearly different.

(e) Present your response clearly.
Threshold segmentation accurately separates the object from the background.

Question 3: Complex Segmentation

(a) Interpret the scenario and explain segmentation challenges.
The image contains multiple regions with overlapping intensity values.

(b) Identify key issues such as intensity overlap.

- Intensity overlap
- Noise
- Touching objects
- Complex textures

(c) Apply an appropriate advanced segmentation method.
Watershed Segmentation

(d) Justify how the method handles complexity.
Watershed separates overlapping and connected objects effectively.

(e) Provide a structured and clear explanation.
Watershed segmentation provides accurate separation of complex regions.

Question 4: Broken Edges

(a) Interpret the scenario and explain why edges are broken.
Detected object boundaries are incomplete due to noise and weak edges.

(b) Identify key issues affecting boundary continuity.

- Noise
- Weak edges
- Missing boundary pixels

(c) Apply a suitable method to improve edge connectivity.
Morphological Closing

(d) Justify your approach logically.
Morphological Closing connects broken edges and fills small gaps.

(e) Present your explanation clearly.
It produces continuous object boundaries for better analysis.

Question 5: Combined Requirement

(a) Interpret the scenario and define the combined requirement.
An industrial inspection system must remove noise while preserving edges for defect detection.

(b) Identify key issues involving noise and edge preservation.

- Image noise
- Edge preservation
- Accurate defect detection

(c) Apply an appropriate sequence of techniques.

1. Median Filter
2. Canny Edge Detection
3. Morphological Closing

(d) Justify the order and selection with strong reasoning.
The Median Filter removes noise, Canny detects important edges, and Morphological Closing connects broken edges, resulting in accurate defect detection.

(e) Present your response in a clear and structured format.
This sequence improves image quality while preserving edges, making defect detection more reliable.